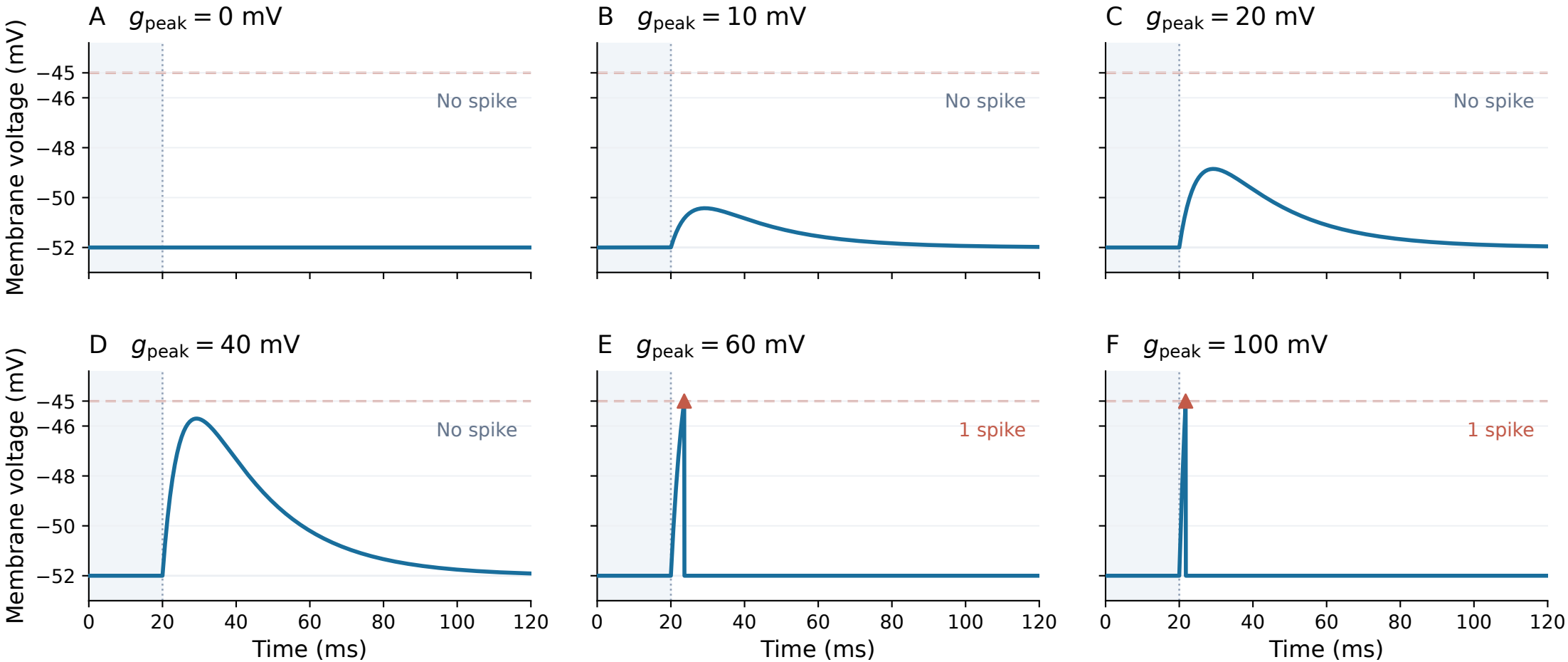
Membrane voltage responses to different synaptic input strengths

Single event at 20 ms: $g \leftarrow g + g_{\text{peak}}$, then $\tau_s\, dg/dt = -g$

$\tau_m\, dV/dt = V_{\text{rest}} - V + g$; $\tau_m = 20$ ms, $\tau_s = 5$ ms, $V_{\text{rest}} = V_{\text{reset}} = -52$ mV



Dashed red: spike threshold (−45 mV) ▲: spike event; voltage resets to −52 mV

At each spike, g resets to 0; refractory period: 2.2 ms. Action-potential waveforms are not modeled.